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Úloha 2.1

$$B = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right)$$

Ak je B ortonormálna báza dim. n , pre bázevé vektory u_i platí

$$\|u_i\|^2 = \langle u_i, u_i \rangle = 1, \quad \forall i = 1, \dots, n$$

$$\langle u_i, u_j \rangle = 0, \quad \forall i, j = 1, \dots, n, \quad i \neq j.$$

Skalárny súčin môžeme v $\mathbb{R}^n$ všeobecne zapísať ako $\langle u, v \rangle = u^T A v$. Maticu A zistíme z rovností, ktoré poznáme z vlastností ortonormálnej bázy:

$$\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\rangle = (1, 2) \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = (a_{11} + 2a_{21}, a_{12} + 2a_{22}) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = a_{11} + 2a_{21} + 2a_{12} + 4a_{22} = 1$$

$$\left\langle \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right\rangle = (3, 7) \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 3 \\ 7 \end{pmatrix} = (3a_{11} + 7a_{21}, 3a_{12} + 7a_{22}) \begin{pmatrix} 3 \\ 7 \end{pmatrix} = 9a_{11} + 21a_{21} + 21a_{12} + 49a_{22} = 1$$

$$\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right\rangle = (a_{11} + 2a_{21}, a_{12} + 2a_{22}) \begin{pmatrix} 3 \\ 7 \end{pmatrix} = 3a_{11} + 6a_{21} + 7a_{12} + 14a_{22} = 0$$

$$\left\langle \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\rangle = (3a_{11} + 7a_{21}, 3a_{12} + 7a_{22}) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 3a_{11} + 7a_{21} + 6a_{12} + 14a_{22} = 0$$

Tým dostávame sústavu lin. rovníc:

$$\left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 9 & 21 & 21 & 49 & 1 \\ 3 & 6 & 7 & 14 & 0 \\ 3 & 7 & 6 & 14 & 0 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 0 & 3 & 3 & 13 & -8 \\ 0 & 0 & 1 & 2 & -3 \\ 0 & 1 & 0 & 2 & -3 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 0 & 0 & 3 & 7 & 1 \\ 0 & 0 & 1 & 2 & -3 \\ 0 & 1 & 0 & 2 & -3 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 0 & 0 & 0 & 1 & 10 \\ 0 & 0 & 1 & 2 & -3 \\ 0 & 1 & 0 & 2 & -3 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 0 & 1 & 0 & 2 & -3 \\ 0 & 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 1 & 10 \end{array} \right)$$

$$a_{11} = 1 - 2a_{12} - 2a_{21} - 4a_{22}$$

$$a_{11} = 53$$

$$a_{12} = -3 - 2a_{22}$$

$$a_{12} = -23$$

$$a_{21} = -3 - 2a_{22}$$

$$a_{21} = -23$$

$$a_{22} = 10$$

$$A = \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix}$$

Z toho dostávame predpis pre skalárny súčin:

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = (x_1, x_2) \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 53x_1y_1 - 23x_2y_1 - 23x_1y_2 + 10x_2y_2$$